Module 6 slides: Repetitive IV Bolus and Intermittent IV Infusions (worked examples)

Module 6 - Multiple Dosings · Lecture slides, worked examples · 23 Sep 2026

Name _____ Date _____

Example 1 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected maximum concentration after the first dose?

Example 1 The same antibiotic (t½ 4 hours, V_D 25% of body weight, 10 mg/kg every 8 hours by IV bolus to a 65-kg female) peaks at 40 mg/L after the first dose. What is the expected minimum concentration after the first dose?

Example 1(a)

An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected maximum concentration at steady state?

Example 1(b)

The same antibiotic (t½ 4 hours, V_D 25% of body weight, 10 mg/kg every 8 hours by IV bolus to a 65-kg female) reaches a maximum of 53.3 mg/L at steady state. What is the expected minimum concentration at steady state?

Example 1(c)

An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected average concentration at steady state?

Example 2 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected concentration 3 hours after injection of the 2nd dose?

Example 3 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female, and the maximum at steady state is 53.3 mg/L. What is the expected concentration 3 hours after injection of the last dose (assuming steady state was attained)?

Slide "Plasma Drug Concentration at Steady State"

An antibiotic with a t½ of 4 hours is at steady state on repeated IV bolus doses, with a maximum of 53.3 mg/L. What is the expected concentration 4 hours after injection of the last dose?

Example 4 A 300-mg dose of an antibiotic is administered as an intravenous (IV) infusion over a period of 2 hours. What is the infusion rate?

Example 4(a)

A 300-mg dose of an antibiotic was administered as an intravenous (IV) infusion over a period of 2 hours ($k = 0.15\ hr^{-1}$, $V_D = 15$ L). What is the plasma drug concentration at the end of the first infusion?

Example 4(b)

A 300-mg dose of an antibiotic was administered as an intravenous (IV) infusion over a period of 2 hours. Six hours after the start of the first infusion, a second dose of 300 mg was infused, again over 2 hours ($k = 0.15\ hr^{-1}$, $V_D = 15$ L). Each infusion reaches 17.28 mg/L at its end. What is the plasma drug concentration 4 hours after the cessation of the second infusion?